Q **July 27, 2026**

The Magical Whip

Andy He

SM-senior 黄瑞轩 is testing a whip. But this whip is brittle and magical: it has *no structural integrity*, so each element $\mathrm{d}m$ of it is in free fall on its own. He holds the whip vertically with its lower end just touching the ground, then lets go.

The whip has mass M and length L, with uniform linear density $\lambda = M/L$. At a later instant a length y of it lies on the ground (piled up at rest), while the remaining length $L - y$ is still falling, the whole falling part descending with speed v as a column.

Assumptions

1. The whip is uniform with linear density $\lambda = M/L$.
2. There is no internal tension: each element is in free fall, and on touching the ground it is instantly brought to rest (perfectly inelastic landing).
3. The pile on the ground is at rest and does not bounce.
4. Air resistance is negligible.

Questions.

Q1. The ground's normal force. Solve for the normal force F_N applied by the ground on the whip as a function of y.